



PROJECT NAME:

Autonomous Detection of Mosquito-breeding Sites And Their Elimination Using An Unmanned Aerial Vehicle

ME 366

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INTRODUCTION

— Every year, thousands of people die from dengue fever. This public health problem is more evident in Bangladesh. In cities, the majority of mosquito-breeding habitats are man-made: bottles, tires, barrels, pots or any area containing stagnant water.

-This project proposes an innovative system to aid in determining mosquito-breeding habitats by employing computer vision on aerial images taken from a UAV and eliminating them.



STEPS

**01. MAKING THE
QUADCOPTER(UAV)**

**02. FLYING THE
QUADCOPTER
ABOVE THE TARGET
LOCATION AND
CAPTURING VIDEO**

**03. DETECTING
BREEDING SITES BY
THE AID OF
COMPUTER VISION**

**04. GETTING THE GPS
COORDINATES OF
THE IDENTIFIED
BREEDING SITES**

**05. ELIMINATION OF
IDENTIFIED BREEDING
SITES**



HARDWARE DETAILS

1. MAIN BODY



2. ESC



3. PROPELLER



4. BATTERY



5. FLIGHT CONTROLLER



6. BLDC MOTOR



7. GPS MODULE



8. CAMERA



9. TRANSMITTER AND RECEIVER



CAD MODEL



APPLICATION DETAILS

For this project, we will be using an **object detection algorithm** to identify some common breeding sites for mosquitoes; such as tires, bottles and stagnant water.

A camera is fitted onto the quadcopter which will record the entire flight while hovering over the targeted location.

The video will then be processed using Python and OpenCV using an object detection algorithm to identify the aforementioned possible breeding sites for mosquitoes and their exact GPS coordinates will be recorded.

Afterwards, the identified breeding sites can be treated in whatever way is best suitable.



Our team

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Thank you

